

Windows Service Persistence Detection and Investigation using Sysmon and Wazuh

Objective

The objective of this simulation was to investigate how Sysmon and Wazuh detect malicious Windows service creation, service-based persistence mechanisms, registry service modifications, and service execution behavior commonly associated with attacker persistence and privilege escalation techniques.

MITRE ATT&CK Mapping

Tactic	Technique	ID
Persistence	Create or Modify System Process: Windows Service	T1543.003
Privilege Escalation	Create or Modify System Process: Windows Service	T1543.003
Defense Evasion	Modify Registry	T1112

Attack Simulation

The following commands were executed in the Windows endpoint lab environment:

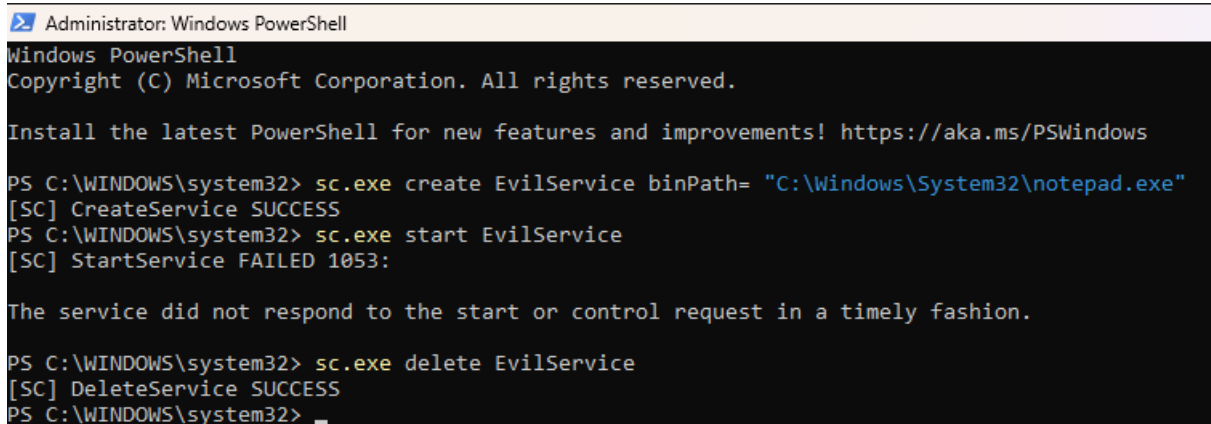
```
sc.exe create EvilService binPath= "C:\Windows\System32\notepad.exe"
```

```
sc.exe start EvilService
```

```
sc.exe delete EvilService
```

Attack Chain Overview

Step	Activity
1	PowerShell launched sc.exe
2	New Windows service created
3	Registry service configuration modified
4	Service execution initiated
5	Payload executed as SYSTEM
6	Service cleanup performed



```
Administrator: Windows PowerShell
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> sc.exe create EvilService binPath= "C:\Windows\System32\notepad.exe"
[SC] CreateService SUCCESS
PS C:\WINDOWS\system32> sc.exe start EvilService
[SC] StartService FAILED 1053:

The service did not respond to the start or control request in a timely fashion.

PS C:\WINDOWS\system32> sc.exe delete EvilService
[SC] DeleteService SUCCESS
PS C:\WINDOWS\system32>
```

Figure 1 — Windows service persistence simulation executed in the lab environment

Sysmon Investigation

Key Sysmon Events Identified

Sysmon Event ID	Detection Purpose
1	Process Creation
13	Registry Value Modification

Additional Windows Events

Log Source	Event ID	Meaning
Service Control Manager	7045	New Service Installed
Service Control Manager	7000	Service Failed to Start
Service Control Manager	7009	Service Timeout

These events formed the primary telemetry chain used during the persistence investigation.

Sysmon Event ID 1 — Service Creation Process Analysis

Key Findings

Field	Value
Event ID	1
Process	sc.exe
Parent Process	powershell.exe
User	socadmin
Integrity Level	High
Command	sc.exe create EvilService

Investigation Analysis

Sysmon Event ID 1 captured execution of `sc.exe` from an elevated PowerShell session to create a new Windows service named:

```
EvilService
```

The parent-child process relationship significantly increased suspicion because attackers commonly use:

- PowerShell

- cmd.exe
- PsExec
- remote administration tools

to establish persistence through malicious service creation.

The following command was observed:

```
sc.exe create EvilService binPath= C:\Windows\System32\notepad.exe
```

Although the payload used during testing was benign, the behavioral pattern strongly resembled real attacker persistence tradecraft.

Important SOC Analyst Note

Legitimate administrators also create Windows services.

Detection engineering should therefore focus on:

- suspicious service names
- unusual binaries
- suspicious parent processes
- abnormal execution timing
- persistence behavior correlation

instead of relying solely on service creation events.

Operational Number of events: 11,915 (1) New events available				
Level	Date and Time	Source	Event ID	Task C...
Information	5/24/2026 11:46:09 AM	Sysmon	13	Registr...
Information	5/24/2026 11:46:09 AM	Sysmon	1	Proces...
Information	5/24/2026 11:41:26 AM	Sysmon	1	Proces...
Information	5/24/2026 11:41:26 AM	Sysmon	1	Proces...
Information	5/24/2026 11:41:12 AM	Sysmon	13	Registr...
Information	5/24/2026 11:41:12 AM	Sysmon	13	Registr...
Information	5/24/2026 11:41:12 AM	Sysmon	1	Proces...
Information	5/24/2026 11:40:34 AM	Sysmon	1	Proces...

Event 1, Sysmon

General

Details

```

ProcessGuid: {04219a00-9080-0a12-c001-000000000000}
ProcessId: 6352
Image: C:\Windows\System32\sc.exe
FileVersion: 10.0.26100.4484 (WinBuild.160101.0800)
Description: Service Control Manager Configuration Tool
Product: Microsoft® Windows® Operating System
Company: Microsoft Corporation
OriginalFileName: sc.exe
CommandLine: "C:\WINDOWS\system32\sc.exe" create EvilService binPath= C:\Windows\System32\notepad.exe
CurrentDirectory: C:\WINDOWS\system32\
User: DESKTOP-HDF09SQ\socadmin
LogonGuid: {842f9a06-883c-6a12-3a02-050000000000}
LogonId: 0x5023A
TerminalSessionId: 1
IntegrityLevel: High
Hashes: MD5=C8F632A326219F85927B3F33A13F7C7D, SHA256=A452BFCB6000D03942DDFADDCA5A7EB121447ED9981BA090446B4A989F1F7AE, IMPHASH=42DADB60BBC453DD1773FDC089E1987
ParentProcessGuid: {842f9a06-964a-6a12-ba01-000000002000}
ParentProcessId: 1472
ParentImage: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
ParentCommandLine: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"
ParentUser: DESKTOP-HDF09SQ\socadmin

```

Log Name: Microsoft-Windows-Sysmon/Operational

Source: Sysmon

Logged: 5/24/2026 11:41:12 AM

Event ID: 1

Task Category: Process Create (rule: ProcessCreate)

Level: Information

Keywords:

User: SYSTEM

Computer: DESKTOP-HDF09SQ

OpCode: Info

Figure 2 — Sysmon Event ID 1 showing sc.exe service creation execution

Sysmon Event ID 13 — Registry Persistence Analysis

Key Findings

Field	Value
Event ID	13
Process	services.exe
Registry Path	HKLM\System\CurrentControlSet\Services\EvilService
Payload	C:\Windows\System32\notepad.exe
User	NT AUTHORITY\SYSTEM

Registry Artifacts Observed

Service Start Configuration

```
HKLM\System\CurrentControlSet\Services\EvilService\Start
```

Service Image Path

```
HKLM\System\CurrentControlSet\Services\EvilService\ImagePath
```

Investigation Analysis

Sysmon Event ID 13 captured registry modifications associated with Windows service persistence configuration.

The Windows Service Control Manager (`services.exe`) modified registry values related to:

- service startup configuration
- service binary path
- persistence execution behavior

The following registry value was observed:

```
ImagePath = C:\Windows\System32\notepad.exe
```

This confirmed successful service persistence creation.

The service startup value observed was:

```
DWORD (0x00000003)
```

This corresponds to:

- Manual/Demand Start service configuration

Important Detection Engineering Insight

Attackers commonly disguise malicious services using fake Microsoft-style names such as:

- WindowsHealthService
- DefenderUpdate
- UpdateService
- svchost_helper

This makes:

- service naming
- payload location
- parent process context

critical detection indicators during SOC investigations.

The screenshot displays the Windows Event Viewer interface. At the top, a status bar indicates 'Operational' and 'Number of events: 11,915 (!) New events available'. Below this is a table of event log entries. The selected entry is highlighted in blue and shows the following details:

Level	Date and Time	Source	Event ID	Task C...
Information	5/24/2026 11:46:09 AM	Sysmon	13	Registr...
Information	5/24/2026 11:46:09 AM	Sysmon	1	Proces...
Information	5/24/2026 11:41:26 AM	Sysmon	1	Proces...
Information	5/24/2026 11:41:26 AM	Sysmon	1	Proces...
Information	5/24/2026 11:41:12 AM	Sysmon	13	Registr...
Information	5/24/2026 11:41:12 AM	Sysmon	13	Registr...
Information	5/24/2026 11:41:12 AM	Sysmon	1	Proces...

The details pane for 'Event 13, Sysmon' is open, showing the 'General' tab. The event data is as follows:

```

Registry value set:
RuleName: T1031,T1050
EventType: SetValue
UtcTime: 2026-05-24 06:11:12.983
ProcessGuid: {842f9a06-8834-6a12-0b00-000000002000}
ProcessId: 880
Image: C:\WINDOWS\system32\services.exe
TargetObject: HKLM\System\CurrentControlSet\Services\EvilService\ImagePath
Details: C:\Windows\System32\notepad.exe
User: NT AUTHORITY\SYSTEM
  
```

At the bottom, a summary of the event is provided:

```

Log Name: Microsoft-Windows-Sysmon/Operational
Source: Sysmon
Event ID: 13
Level: Information
User: SYSTEM
OpCode: Info
Logged: 5/24/2026 11:41:12 AM
Task Category: Registry value set (rule: RegistryEvent
Keywords:
Computer: DESKTOP-HDF09SQ
  
```

Figure 3 — Sysmon Event ID 13 showing Windows service registry persistence modifications

Service Execution Investigation

Sysmon Event ID 1 — Payload Execution Analysis

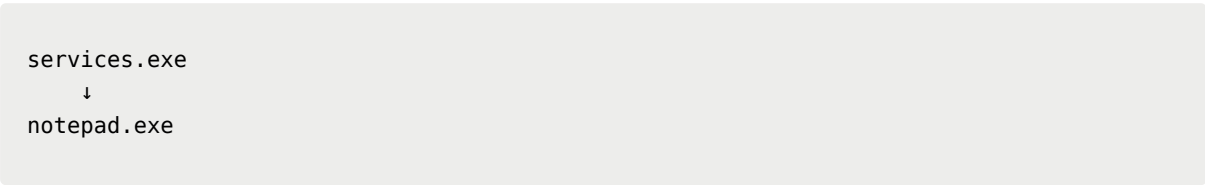
Key Findings

Field	Value
Event ID	1
Process	notepad.exe
Parent Process	services.exe
User	NT AUTHORITY\SYSTEM
Integrity Level	System

Investigation Analysis

Following service startup activity, Sysmon Event ID 1 captured execution of the configured payload through the Windows Service Control Manager.

The following parent-child relationship was observed:



This confirmed successful service execution behavior.

The process executed under:

NT AUTHORITY\SYSTEM

which significantly increased severity because service-based persistence commonly executes with elevated privileges.

Important SOC Analyst Observation

The payload used during testing:

- was benign
- did not function as a true service binary

As a result, additional Service Control Manager errors were generated:

- Event ID 7000
- Event ID 7009

This behavior realistically demonstrated how malformed or non-service executables can still generate valuable telemetry during persistence investigations.

Operational Number of events: 11,915 (!) New events available				
Level	Date and Time	Source	Event ID	Task C...
Information	5/24/2026 11:46:09 AM	Sysmon	13	Registr...
Information	5/24/2026 11:46:09 AM	Sysmon	1	Proces...
Information	5/24/2026 11:41:26 AM	Sysmon	1	Proces...
Information	5/24/2026 11:41:26 AM	Sysmon	1	Proces...

Event 1, Sysmon	
General	Details
Process Create: RuleName: - UtcTime: 2026-05-24 06:11:26.651 ProcessGuid: {842f9a06-968e-6a12-c201-000000002000} ProcessId: 9752 Image: C:\Windows\System32\notepad.exe FileVersion: 10.0.26100.7309 (WinBuild.160101.0800) Description: Notepad Product: Microsoft® Windows® Operating System Company: Microsoft Corporation OriginalFileName: NOTEPAD.EXE CommandLine: C:\Windows\System32\notepad.exe CurrentDirectory: C:\WINDOWS\system32\ User: NT AUTHORITY\SYSTEM LogonGuid: {842f9a06-8834-6a12-e703-000000000000} LogonId: 0x3E7 TerminalSessionId: 0 IntegrityLevel: System Hashes: MD5=9E60393DA455F93B0EC32CF124432651,SHA256=84B484FD3636F2CA3E468D2821D97AACDE8A143A2724A3AE65F48A33CA2FD258,IMPHASH=0E6BCCF88F4251909D1746DBA78CBA57 ParentProcessGuid: {842f9a06-8834-6a12-0b00-000000002000} ParentProcessId: 880 ParentImage: C:\Windows\System32\services.exe ParentCommandLine: C:\WINDOWS\system32\services.exe ParentUser: NT AUTHORITY\SYSTEM	
Log Name:	Microsoft-Windows-Sysmon/Operational
Source:	Sysmon
Event ID:	1
Level:	Information
User:	SYSTEM
OpCode:	Info
Logged:	5/24/2026 11:41:26 AM
Task Category:	Process Create (rule: ProcessCreate)
Keywords:	
Computer:	DESKTOP-HDF09SQ

Figure 4 — Sysmon Event ID 1 showing service payload execution under SYSTEM context

Service Control Manager Investigation

Windows Event ID 7045 — Service Installation Analysis

Key Findings

Field	Value
Event ID	7045
Service Name	EvilService
Image Path	C:\Windows\System32\notepad.exe
Service Account	LocalSystem
Start Type	Demand Start

Investigation Analysis

Windows Event ID 7045 confirmed successful installation of a new Windows service within the operating system.

This event provided additional high-value telemetry including:

- service name
- payload path
- startup configuration
- execution account

The following service account was observed:

```
LocalSystem
```

Execution under LocalSystem significantly increases investigation severity because attackers frequently abuse Windows services for:

- persistence
- privilege escalation
- stealth execution

Operational

Number of events: 11,915 (0) New events available

Level	Date and Time	Source	Event ID	Task C...
 Information	5/24/2026 11:41:26 AM	Sysmon	1	Proces...
 Information	5/24/2026 11:41:26 AM	Sysmon	1	Proces...
 Information	5/24/2026 11:41:12 AM	Sysmon	13	Registr...
 Information	5/24/2026 11:41:12 AM	Sysmon	13	Registr...

Event 1, Sysmon

General

Details

Process Create:
RuleName: -
UtcTime: 2026-05-24 06:11:26.612
ProcessGuid: {842f9a06-968e-6a12-c101-000000002000}
ProcessId: 7044
Image: C:\Windows\System32\sc.exe
FileVersion: 10.0.26100.4484 (WinBuild.160101.0800)
Description: Service Control Manager Configuration Tool
Product: Microsoft® Windows® Operating System
Company: Microsoft Corporation
OriginalFileName: sc.exe
CommandLine: "C:\WINDOWS\system32\sc.exe" start EvilService
CurrentDirectory: C:\WINDOWS\system32\
User: DESKTOP-HDF09SQ\socadmin
LogonGuid: {842f9a06-883c-6a12-3a02-050000000000}
LogonId: 0x5023A
TerminalSessionId: 1
IntegrityLevel: High
Hashes: MD5=C8F632A326219F85927B3F33A13F7C7D,SHA256=A452BFCB6000D03942DDFADDCA5A7EB121447ED99B1BA090446B4A989F1F7AE,IMPHASH=42DADB60B8BC453DD1773FDC089E19B7
ParentProcessGuid: {842f9a06-964a-6a12-ba01-000000002000}
ParentProcessId: 1472
ParentImage: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
ParentCommandLine: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"
ParentUser: DESKTOP-HDF09SQ\socadmin

Log Name: Microsoft-Windows-Sysmon/Operational

Source: Sysmon

Logged: 5/24/2026 11:41:26 AM

Event ID: 1

Task Category: Process Create (rule: ProcessCreate)

Level: Information

Keywords:

User: SYSTEM

Computer: DESKTOP-HDF09SQ

OpCode: Info

Figure 5 — Windows Event ID 7045 confirming new Windows service installation

Wazuh Detection Analysis

Wazuh Rules Triggered

Rule ID	Severity	Description
92307	Medium	Evidence of new service creation in registry
61138	Medium	New Windows Service Created
61102	Medium	Windows System Error Event
598	Medium	Registry Key Entry Added
594	Medium	Registry Key Integrity Checksum Changed
750	Medium	Registry Value Integrity Checksum Changed

Investigation Analysis

Wazuh successfully detected multiple stages of the Windows service persistence lifecycle through:

- Sysmon telemetry
- Windows System logs
- registry integrity monitoring
- File Integrity Monitoring (FIM)

The strongest detection indicators included:

- creation of a new service
- registry service modifications
- service execution behavior
- SYSTEM-level process execution
- persistence-related registry changes

Wazuh correctly mapped the activity to:

Technique	ID
Windows Service Persistence	T1543.003

Important False Positive Analysis

Legitimate software installers and administrators frequently create Windows services.

However, the following characteristics significantly increased suspicion during this investigation:

- PowerShell launching sc.exe
- suspicious service name
- SYSTEM-level execution
- unusual payload behavior
- registry persistence activity
- non-standard service executable

This highlights the importance of behavioral correlation instead of relying solely on isolated service creation events.

```
> May 24, 2026 @ 11:41:13.056 input.type: log agent.ip: 192.168.56.106 agent.name: DESKTOP-HDF09SQ agent.id: 001 manager.name: wazuh-server
data.win.eventdata.image: C:\\WINDOWS\\system32\\services.exe data.win.eventdata.targetObject: HKLM\\System\\CurrentControlSet
\\Services\\EvilService\\ImagePath data.win.eventdata.processGuid: {842f9a06-8834-6a12-0b00-000000002000}
data.win.eventdata.processId: 880 data.win.eventdata.utcTime: 2026-05-24 06:11:12.983 data.win.eventdata.ruleName: T1031,T105
0 data.win.eventdata.details: C:\\Windows\\System32\\notepad.exe data.win.eventdata.eventType: SetValue
```

Figure 6 — Wazuh detection for Windows service registry persistence activity

```
> May 24, 2026 @ 11:41:13.096 input.type: log agent.ip: 192.168.56.106 agent.name: DESKTOP-HDF09SQ agent.id: 001 manager.name: wazuh-server
data.win.eventdata.serviceType: user mode service data.win.eventdata.accountName: LocalSystem
data.win.eventdata.imagePath: C:\\Windows\\System32\\notepad.exe data.win.eventdata.startType: demand start
data.win.eventdata.serviceName: EvilService data.win.system.eventID: 7045 data.win.system.eventSourceName: Service Control Ma
nager data.win.system.keywords: 0x8080000000000000 data.win.system.providerGuid: {555908d1-a6d7-4695-8e1e-26931d2012f4}
```

Figure 7 — Wazuh detection confirming new Windows service installation

```
{
  "_index": "wazuh-alerts-4.x-2026.05.24",
  "_id": "yc6bWJ4Bsdig1Cmu8MJD",
  "_version": 1,
  "_score": null,
  "_source": {
    "input": {
      "type": "log"
    },
    "agent": {
      "ip": "192.168.56.106",
      "name": "DESKTOP-HDF09SQ",
      "id": "001"
    },
    "manager": {
      "name": "wazuh-server"
    },
    "data": {
      "win": {
        "eventdata": {
          "image": "C:\\\\WINDOWS\\\\system32\\\\services.exe",
          "targetObject":
            "HKLM\\\\System\\\\CurrentControlSet\\\\Services\\\\EvilService\\\\ImagePath",
```

```

"processGuid": "{842f9a06-8834-6a12-0b00-000000002000}",
"processId": "880",
"utcTime": "2026-05-24 06:11:12.983",
"ruleName": "T1031,T1050",
"details": "C:\\\\Windows\\\\System32\\\\notepad.exe",
"eventType": "SetValue",
"user": "NT AUTHORITY\\\\SYSTEM"
},
"system": {
"eventID": "13",
"keywords": "0x8000000000000000",
"providerGuid": "{5770385f-c22a-43e0-bf4c-06f5698ffbd9}",
"level": "4",
"channel": "Microsoft-Windows-Sysmon/Operational",
"opcode": "0",
"message": "\"Registry value set:\\r\\nRuleName: T1031,T1050\\r\\nEventType:
SetValue\\r\\nUtcTime: 2026-05-24 06:11:12.983\\r\\nProcessGuid: {842f9a06-
8834-6a12-0b00-000000002000}\\r\\nProcessId: 880\\r\\nImage:
C:\\\\WINDOWS\\\\system32\\\\services.exe\\r\\nTargetObject:
HKLM\\\\System\\\\CurrentControlSet\\\\Services\\\\EvilService\\\\ImagePath\\r\\nDetails:
C:\\\\Windows\\\\System32\\\\notepad.exe\\r\\nUser: NT AUTHORITY\\\\SYSTEM\"",
"version": "2",
"systemTime": "2026-05-24T06:11:12.9837170Z",
"eventRecordID": "17196",
"threadID": "4088",
"computer": "DESKTOP-HDF09SQ",
"task": "13",
"processID": "3420",
"severityValue": "INFORMATION",
"providerName": "Microsoft-Windows-Sysmon"
}
},
"rule": {
"firedtimes": 1,
"mail": false,
"level": 3,
"description": "Evidence of new service creation found in registry under

```

```
HKLM\\System\\CurrentControlSet\\Services\\EvilService\\ImagePath
binary is: C:\\Windows\\System32\\notepad.exe",
"groups": [
  "sysmon",
  "sysmon_eid13_detections",
  "windows"
],
"mitre": {
  "technique": [
    "Windows Service"
  ],
  "id": [
    "T1543.003"
  ],
  "tactic": [
    "Persistence",
    "Privilege Escalation"
  ]
},
"id": "92307",
"location": "EventChannel",
"decoder": {
  "name": "windows_eventchannel"
},
"id": "1779603073.2093721",
"timestamp": "2026-05-24T06:11:13.056+0000"
},
"fields": {
  "timestamp": [
    "2026-05-24T06:11:13.056Z"
  ]
},
"sort": [
  1779603073056
]
}
```

```
{
  "_index": "wazuh-alerts-4.x-2026.05.24",
  "_id": "ys6bWJ4Bsdig1Cmu8MJD",
  "_version": 1,
  "_score": null,
  "_source": {
    "input": {
      "type": "log"
    },
    "agent": {
      "ip": "192.168.56.106",
      "name": "DESKTOP-HDF09SQ",
      "id": "001"
    },
    "manager": {
      "name": "wazuh-server"
    },
    "data": {
      "win": {
        "eventdata": {
          "serviceType": "user mode service",
          "accountName": "LocalSystem",
          "imagePath": "C:\\\\Windows\\\\System32\\\\notepad.exe",
          "startType": "demand start",
          "serviceName": "EvilService"
        },
        "system": {
          "eventID": "7045",
          "eventSourceName": "Service Control Manager",
          "keywords": "0x8080000000000000",
          "providerGuid": "{555908d1-a6d7-4695-8e1e-26931d2012f4}",
          "level": "4",
          "channel": "System",
          "opcode": "0",
          "message": "\"A service was installed in the system.\\r\\n\\r\\nService Name:
EvilService\\r\\nService File Name:
C:\\\\Windows\\\\System32\\\\notepad.exe\\r\\nService Type: user mode
service\\r\\nService Start Type: demand start\\r\\nService Account:"

```



```
LocalSystem\\",
"version": "0",
"systemTime": "2026-05-24T06:11:12.9834392Z",
"eventRecordID": "3508",
"threadID": "2244",
"computer": "DESKTOP-HDF09SQ",
"task": "0",
"processID": "880",
"severityValue": "INFORMATION",
"providerName": "Service Control Manager"
}
},
"rule": {
"firedtimes": 1,
"mail": false,
"level": 5,
"description": "New Windows Service Created",
"groups": [
"windows",
"windows_system"
],
"mitre": {
"technique": [
"Windows Service"
],
"id": [
"T1543.003"
],
"tactic": [
"Persistence",
"Privilege Escalation"
]
},
"id": "61138"
},
"location": "EventChannel",
"decoder": {
```

```

"name": "windows_eventchannel"
},
"id": "1779603073.2096642",
"timestamp": "2026-05-24T06:11:13.096+0000"
},
"fields": {
"timestamp": [
"2026-05-24T06:11:13.096Z"
]
},
"sort": [
1779603073096
]
}

```

Figure 8 — Detailed Wazuh telemetry for Windows service persistence behavior

Attack Timeline

Time	Event
11:41:12 AM	sc.exe create EvilService executed
11:41:12 AM	Sysmon Event ID 1 generated for service creation
11:41:12 AM	Sysmon Event ID 13 generated for registry service modifications
11:41:12 AM	Windows Event ID 7045 generated
11:41:13 AM	Wazuh generated service persistence alerts
11:41:26 AM	sc.exe start EvilService executed
11:41:26 AM	notepad.exe launched by services.exe
11:41:56 AM	Service Control Manager generated timeout errors
11:46:09 AM	sc.exe delete EvilService executed
11:46:09 AM	Registry cleanup modifications detected

Key Detection Findings

Finding	Impact
PowerShell launching sc.exe	Suspicious persistence activity
New Windows service created	High-value persistence indicator
Registry service modifications	Persistence confirmation
SYSTEM-level payload execution	Increased severity
services.exe spawning payload	Service execution validation
Multiple telemetry sources correlated	Improved detection confidence
Registry integrity changes detected	Additional persistence visibility

Primary Investigation Artifacts

Sysmon Event IDs

Event ID	Meaning
1	Process Creation
13	Registry Value Modification

Windows Event IDs

Event ID	Meaning
7045	New Service Installed
7000	Service Failed to Start
7009	Service Timeout

Wazuh Rule IDs

Rule ID	Purpose
92307	New service registry persistence
61138	New Windows service created
61102	Windows service errors
598	Registry key added

Rule ID	Purpose
594	Registry key modified
750	Registry value modified

Conclusion

This simulation demonstrated how Sysmon and Wazuh provide strong visibility into Windows service persistence techniques commonly associated with attacker persistence and privilege escalation behavior.

Behavioral correlation between:

- PowerShell execution
- sc.exe activity
- service registry modifications
- Windows Service Control Manager events
- SYSTEM-level payload execution
- registry integrity monitoring

significantly improved detection confidence and investigation quality.

The investigation also highlighted the importance of multi-source telemetry correlation because modern SOC investigations rely on combined visibility from:

- Sysmon
- Windows Event Logs
- registry monitoring
- SIEM correlation
- persistence telemetry

rather than isolated security events alone.